Dust Particles and Gas+Dust Self-Gravity in the AthenaK Shearing Box: I. The Dust-Module Merge (Phase 4a), the Shear-Periodic Deposit Fold, the Ideal-Gas Energy Update and the Particle Timestep under Shear (Phase 4b, i–iii), Dust Self-Gravity (Phase 4c), and the Vertical Policy, Restart and Diagnostics (Phase 4d)

`athenak-multigrid` tree, branch `dust-multigrid`

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Abstract

This document opens the *dust track* of the self-gravity program: Lagrangian dust particles in the shearing-box code with multigrid self-gravity on refined meshes, toward the gas+dust gravito-turbulent boxes of Baehr, Zhu & Yang (2022) with the radial pressure gradient they omitted. It records what the code does on test problems, phase by phase; the mechanisms are in the companion `dust_selfgravity_implementation.pdf`.

Phase 4a merges the sibling fork `athenak-dust` — particle-mesh dust with Benítez-Llambay–Krapp–Pessah implicit drag inside the Krapp et al. (2024) IMEX integrator, validated in its own repository — into the multigrid tree, and asks two questions: is the dust module unchanged by the merge, and is the gravity machinery unchanged by the dust module? Both answers are yes to every printed digit. On the merged tree the dust fork's own acceptance numbers reproduce exactly: the N-species damping ladder converges at second order in Δt with the reference errors 1.3529×10^{-7}, 3.4627×10^{-8}, 8.7724×10^{-9}, 2.2075×10^{-9} and total momentum conserved to 10^{-16}; the multi-species Nakagawa–Sekiya–Hayashi drift equilibrium holds to 10^{-16} over ten orbits; the damped dusty sound wave converges toward second order in resolution; the ion-neutral C-shock under `imex2+` is bit-identical to its baseline. The gravity solver's documented statics reproduce exactly (slab sheet error 1.203×10^{-12}, Tomida & Stone contraction 0.125, boundary-annuli shearing wave 5.950×10^{-3} / 2.057×10^{-3}), bit-identically at 4 ranks. The one defect found — a multi-rank deadlock of the additive deposit exchange against the tree's rank-packed messaging — is fixed and ranks 1, 2 and 4 agree to all digits. Finally the gas-only stratified gravito-turbulence box evolved with `rk2` and with `imex2+` (the integrator the dust module requires) agrees in every history column to between 2×10^{-6} and 5×10^{-3} at $t = 3\,\Omega^{-1}$, the expected spread between two second-order integrators.

A **second merge** the same day brought in the dust fork's latest seven commits. The whole battery reproduces on the result to every digit (the gas-only run bit for bit), and the fork's new predictor–corrector coupling under plain `rk2` (`coupling = pc2`) is exercised for the first time here: second order in the damping ladder (orders 2.23, 2.10, 2.04) and in the dusty wave (1.85, 1.98, 2.00), NSH to round-off, rank-independent at 1, 2 and 4 ranks — the integrator freedom that lets dust run under the same `rk2` as every gravity result before it.

Phase 4b (i) adds the conservative azimuthal remap of ghost deposits across the shear-periodic radial faces, the first of the gaps the dust module carried. A particle-free unit test

certifies the new fold: the plain exchange delivers nothing across the faces, the fold is exact at integer shifts (3.5×10^{-16}) including the overlap multiplicity of corner rows, mass is conserved to 10^{-15}, nothing leaks outside the receiving strips, the remap error converges at second order (donor cell), second/third (PLM), third (PPMX) in a fixed-phase scan, and 1, 2 and 4 ranks agree to every digit. The 3D shear-periodic NSH equilibrium still holds to round-off; a deterministic azimuthally structured dust distribution with back-reaction agrees between serial and 4-rank runs to 10^{-14} over an orbit, and shows the old unsheared exchange was a 1–7% error in the gas kinetic energy while the remap order matters only at 10^{-7}.

Phase 4b (ii) removes the isothermal restriction on back-reaction. With every drag kick booking its kinetic-energy change, the ideal-gas damping test keeps the momentum dynamics identical to the isothermal reference and changes the internal energy by 1% (IMEX) / 2% (PC2) of the dissipated energy at a step equal to the stiffest stopping time — the low-storage RK combination gap, converging with the step; the frictional-heating option conserves $E_{\mathrm{gas}} + K_{\mathrm{dust}}$ to 4–6% of the dissipated energy at that step, first order in Δt; and on the 3D shear-periodic NSH state the heating rate matches the analytic dissipation to 1.5%, 0.7%, 0.4% as the step is halved twice, with the equilibrium itself at round-off in both modes.

Phase 4b (iii) drops the background shear from the particle transport limit and replaces it by a MeshBlock-crossing guard on the total velocity. In an $L_x = 8$ shear-periodic box the ten-orbit NSH run takes 4588 cycles instead of 24043 (5.2×; the step is now the gas limit), at round-off in every drift velocity, serially, at 4 ranks and under PC2; the guard reproduces its formula to four digits when its safety fraction is varied, and with thin MeshBlocks it keeps a run clean that aborts on the first cycle without it. A structured run is rank-independent to 10^{-14} at the larger step. The scan the larger step made necessary shows the IMEX coupling's own truncation error at $\Delta t/t_{s,\min} = 1.4$: 1% in the gas kinetic energy in the quiet phase, 26% during the run's streaming burst, converging at second order — an error the legacy limit had been hiding in wide boxes by forcing $\Delta t \approx t_{s,\min}/4$ — while PC2 at its `rk2` step sits within 2% of the converged value.

Phase 4c puts the dust into the Poisson equation and the potential onto the particles. A *twin* construction certifies the source: with one particle per cell carrying a fraction of a known density profile and NGP deposits, the multigrid and FFT solves reproduce the gas-only Tomida–Stone error (3.014475×10^{-5}) and the slab-open sheet error (1.203474×10^{-12}) to every digit, serially and on 4 ranks, IMEX and PC2; the force gathered at the particles converges to the analytic acceleration at second order (NGP 2.0×10^{-4}, TSC 2.0×10^{-3} at 128^3); and the net force on gas plus dust vanishes to 10^{-13} for both kernels, the momentum antisymmetry the shared-kernel design guarantees. Dynamically, test particles in a self-gravitating isothermal slab oscillate with the exact anharmonic period to 5×10^{-5}, the shear-periodic NSH equilibrium holds to round-off with the dust in the source, and a structured gravitating run is rank-independent to 10^{-14}. Three findings are recorded: an uninitialized upstream particle timestep (fixed), the multigrid's need for cubic cells and multi-block periodic directions, and a numerical instability in 3D boxes at $c_s \Delta t/\Delta z \gtrsim 0.35$, at first attributed to the TSC drag coupling.

Phase 4d identifies that instability: its eigenmode is the 3D grid checkerboard of the gas, and a gas-only run seeded with 10^{-10} noise grows it at `cfl` 0.45 and 0.40 under both integrators and damps it at 0.30 — the shearing-box hydro's own Courant limit on cubic cells, which the dust merely seeded. Particles crossing the vertical faces are removed with their mass accounted (64 of 64 in the escape test, serially and on 2 ranks), particle restart reproduces an uninterrupted run to every digit (serially and on 4 ranks), and the new `dust_dpm` output reproduces the twin profile exactly and the uniform lattice to round-off on 4 ranks.

Contents

1 Plan and scope

1.1 The program

The science target is the interplay of gas gravitational instability and dust concentration: 3D stratified, cooling, gravito-turbulent shearing boxes with superparticles (Baehr, Zhu & Yang 2022: $512^2 \times 256$, $\beta = 10$ cooling with an irradiation floor, $\gamma = 5/3$, $\mathrm{St} = 0.1$, 1, 10, $Z = 0.01$, $Q_0 = 1.02$), extended by the radial pressure gradient that admits the streaming instability, and eventually adaptive zoom on the dust clumps — the configuration their limitations section names. The route (implementation document, §1.2): Phase 4a merges the dust module; 4b adds what the 3D shearing box needs from it (shear-remapped additive deposits, ideal-gas back-reaction, a relaxed particle timestep); 4c adds dust self-gravity through the existing Poisson solve; 4d builds the science configuration; Phase 5 takes particles onto refined meshes.

1.2 What Phase 4a must establish

1. **Dust fidelity.** The dust fork ships its validation in `../athenak_dust_tests/` (notebooks with printed reference values). On the merged tree, those references must reproduce: the damping ladder (second order in Δt, momentum to round-off), the NSH drift equilibrium (round-off over ten orbits), the damped dusty wave (spatial convergence), the random-placement and single-block layout checks, and the regression of the ion-neutral C-shock that shares the `imex2+` coefficient code.
2. **Gravity regression.** Three static multigrid tests whose numbers are documented in the multigrid validation document: the slab-open sheet test, the Tomida & Stone §4.1 contraction, and the boundary-annuli shearing wave including its zero-phase bit-identity with the periodic twin.
3. **Rank independence.** Dust and gravity at 1, 2 and 4 MPI ranks.
4. **Integrator.** The dust module's IMEX coupling requires `imex2+`; all gravity validation so far used `rk2`. A gas-only stratified gravito-turbulence box under both integrators, compared column by column.
5. **The second merge** (§4). After catching up with the dust fork: the battery again, and the new PC2/hybrid couplings under `rk2`.
6. **Phase 4b (i)** (§5). The shear-periodic deposit fold: exactness, convergence, conservation, rank independence, and its effect on a structured 3D run.
7. **Phase 4b (ii)** (§6). The ideal-gas energy update: momentum dynamics unchanged, energy budgets of both modes versus the step, the heating rate against the analytic dissipation.

2 Test setup

2.1 Builds

Serial and MPI builds in the tree's `build/` and `build-mpi/`, both with `-D PROBLEM=built_in_pgens -D Athena_ENABLE_FFT=ON`, the MPI one with `-D Athena_ENABLE_MPI=ON` (Open MPI via Homebrew). Everything below was first run on a scratch clone at the same commits and then rerun with the tree's own binaries; the two sets of results are identical, including the gravito-turbulence histories bit for bit.

2.2 Inputs and commands

Dust inputs come from the dust-tests repository (paths relative to the directory holding both trees):

```
# damping ladder (tlim = 1, so dt = 1/ncycle)
for c in 0.4 0.2 0.1 0.05; do
  athena -i athenak_dust_tests/inputs/dust_damping.athinput particles/ppc=3 time/
    cfl_number=$c
done
# layout variants
athena -i .../dust_damping.athinput particles/ppc=4 problem/lattice=false
athena -i .../dust_damping.athinput particles/ppc=4 problem/lattice=false dust/deposit=
    ngp
athena -i .../dust_damping.athinput particles/ppc=3 meshblock/nx1=32 meshblock/nx2=32
# NSH equilibrium, dusty wave, C-shock regression
athena -i .../dust_nsh.athinput
athena -i .../dusty_wave.athinput mesh/nx1=$n meshblock/nx1=$((n/2))     # n = 32..256
athena -i inputs/ion-neutral/perp_cshock.athinput time/integrator=imex2+ time/nlim=500
# rank independence (drifting particles cross MeshBlock and rank boundaries)
mpirun -np $np athena -i .../dust_damping.athinput particles/ppc=3 problem/v0=0.5 dust/
    taus_3=0.5
# gravity statics
athena -i inputs/tests/mg_poisson_slab.athinput
athena -i inputs/tests/ts41_sin3.athinput
athena -i inputs/tests/mg_poisson_shear_bndry.athinput problem/profile=shwave problem/
    time0=0.37
# integrator twin (gas only, uniform grid)
athena -i inputs/shearing_box/gravito_turb_amr_test.athinput mesh_refinement/refinement=
    none \
       time/tlim=3.0 [time/integrator=imex2+ time/cfl_number=0.3]
```

Phase 4b (iii) (§7; the wide box is the tracked 3D input with $x_1 \in [-4, 4]$ at the same cell size):

```
WIDE="mesh/nx1=64 mesh/x1min=-4.0 mesh/x1max=4.0 meshblock/nx1=32"
IN=inputs/dust/dust_nsh_shear3d.athinput
athena -i $IN $WIDE dust/dt_transport=full            # legacy limit, 10 orbits
athena -i $IN $WIDE dust/dt_transport=relative        # default; also mpirun -np 4, and
                                                      # dust/coupling=pc2 time/integrator
    =rk2 time/cfl_number=0.3
# structured state, one orbit, both limits, serial and 4 ranks, IMEX and PC2
athena -i $IN $WIDE problem/mass_mod_amp=0.5 problem/check_equilibrium=false \
       time/tlim=6.283185307179586 dust/dt_transport={full,relative}
# guard scaling (three cycles) and the thin-block abort demonstration (one orbit)
athena -i $IN $WIDE dust/dt_block_safety={0.1,0.2,0.4} time/nlim=3 time/ndiag=1
athena -i $IN $WIDE meshblock/nx2=4 dust/dt_block_safety={0.5,4.0} time/tlim
    =6.283185307179586
```

Phase 4d (§9):

```
# the instability: gas only, seeded (the tracked 3D NSH input without its <particles>/<
    dust> blocks)
athena -i nsh_gasonly.athinput mesh/nx3=32 meshblock/nx3=32 problem/gas_vpert=1e-10 \
       problem/check_equilibrium=false time/cfl_number={0.45,0.40,0.30} [time/integrator=
    rk2]
# escape, restart, dust_dpm
```

```
athena -i inputs/dust/dust_grav_orbit.athinput problem/vz0=0.3 dust/gravity_force=false
    time/tlim=3
athena -i nsh_rst.athinput; athena -r nsh3d.00001.rst          # structured NSH, dumps
    every 3.0
athena -i <input with an <output> block variable = dust_dpm>
# the two-stage science setup (mini box): gas alone, then restart + insert
athena -i inputs/shearing_box/gravito_turb_baehr.athinput <mesh overrides> hydro_srcterms
    /beta_cooling=false \
       time/tlim=0.5 output4/dt=0.25
athena -r s1/rst/gtb.00001.rst -i inputs/shearing_box/gravito_turb_baehr_dust.athinput
    dust/back_reaction={true,false}
```

Phase 4c (§8):

```
# static twins (nlim = 0): gas-only reference, then the dust carrying the profile
athena -i inputs/dust/dust_poisson_twin.athinput problem/dust_frac=0.0 dust/gravity=false
athena -i inputs/dust/dust_poisson_twin.athinput problem/dust_frac={1.0,0.5} dust/deposit
    ={ngp,tsc} \
       [gravity/solver=fft] [dust/coupling=pc2 time/integrator=rk2] [mesh/nx*=32|128
    meshblock/nx*=16|32]
athena -i inputs/dust/dust_poisson_twin.athinput problem/profile=modes problem/dust_part=
    blob   # momentum
athena -i inputs/dust/dust_poisson_twin_slab.athinput problem/dust_frac={0.0,0.5,1.0} [
    problem/dust_gas_floor=0]
# test particles in the self-gravitating slab, five periods; scan time/dtmax
athena -i inputs/dust/dust_grav_orbit.athinput [time/dtmax=0.05|0.025|0.0125] [dust/
    coupling=pc2 ...]
# NSH with gravity: uniform lattice (equilibrium), structured state at rest (four_pi_G =
    2)
athena -i inputs/dust/dust_nsh_shear3d_grav.athinput time/tlim=6.283185307179586
athena -i inputs/dust/dust_nsh_shear3d_grav.athinput time/tlim=6.283185307179586 problem/
    mass_mod_amp=0.5 \
       problem/check_equilibrium=false problem/etavk=0.0 hydro_srcterms/const_accel_val
    =0.0 gravity/four_pi_G=2.0
```

Each dust problem generator writes a `<basename>-errs.dat` line per run; the gravity statics print their metrics to standard output. All result files are archived under `validation/run/dust4a/` (see its `README.txt`), which the figure scripts `figures/plot_dust4a_*.jl` read.

3 Results

3.1 Dust fidelity

Damping ladder (Krapp et al. 2024 §3.1). Uniform gas at rest coupled to three dust species, $t_s = (0.01, 0.1, 1.0)$, $\epsilon_s = 1/3$ each, one particle of every species at every cell center so the discrete update per cell *is* the integrator's map of the ODE system. Errors are against a fine-step RK4 reference; the first run has $\Delta t > t_{s,1}$ (stiff regime).

Δt	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\mathrm{tot}}\rvert$
1.235×10^{-2}	1.3529×10^{-7}	1.3711×10^{-7}	1.4641×10^{-7}	6.8939×10^{-7}	1.6×10^{-16}
6.211×10^{-3}	3.4627×10^{-8}	3.5096×10^{-8}	3.7325×10^{-8}	1.7630×10^{-7}	7.5×10^{-17}
3.115×10^{-3}	8.7724×10^{-9}	8.8920×10^{-9}	9.4213×10^{-9}	4.4631×10^{-8}	2.0×10^{-16}
1.558×10^{-3}	2.2075×10^{-9}	2.2377×10^{-9}	2.3649×10^{-9}	1.1225×10^{-8}	1.7×10^{-16}
order of err_u	1.98, 1.99, 1.99				
reference (dust tests, 2026-07-14)	$\mathrm{err}_u = 1.3529 \times 10^{-7}$, 3.4627×10^{-8}, 8.7724×10^{-9}, 2.2075×10^{-9}; $\lvert\Delta P\rvert \sim 10^{-16}$				

Every reference digit reproduces (Fig. 1). The layout variants: random positions with TSC and with NGP deposit give $\mathrm{err}_u = 5.29 \times 10^{-6}$ and 5.77×10^{-6} (the 10^{-5}-level departure from the homogeneous ODE is physical: density fluctuations couple to pressure) with momentum to 3.8×10^{-16}; the single-MeshBlock lattice run reproduces the four-block ladder row to all digits, which is the halo-exchange exactness check.

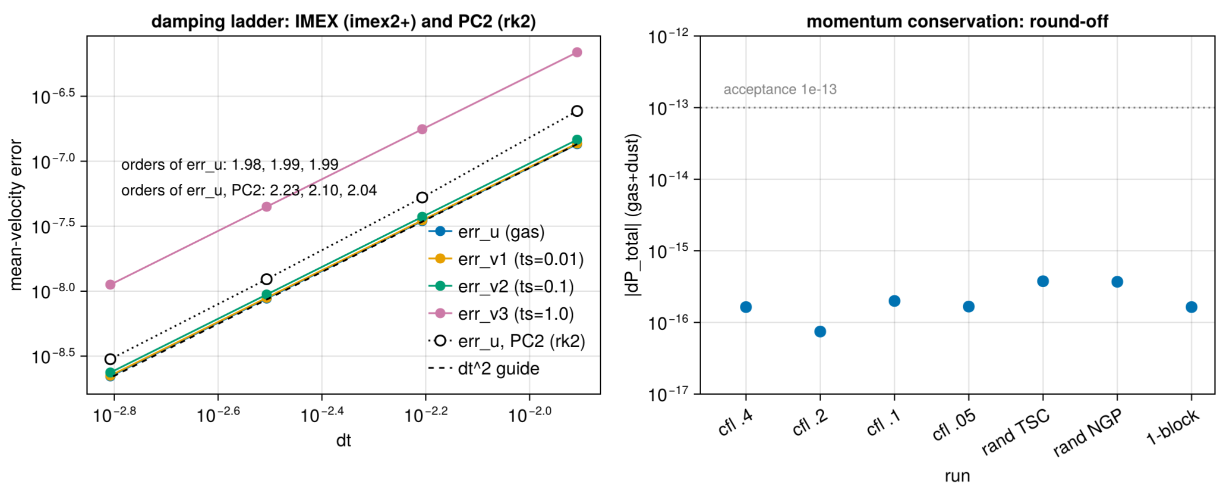

Figure 1: The damping ladder on the merged tree: second-order convergence of all four mean-velocity errors in Δt under the IMEX coupling (filled), the gas error under the PC2 coupling with `rk2` after the second merge (open, §4), and the total gas+dust momentum change of every IMEX run against the 10^{-13} acceptance line (right).

NSH drift equilibrium (2D r–z, ten orbits). Gas plus three species ($t_s = 0.01, 0.1, 1.0$; $\epsilon = 0.2, 0.5, 0.3$) started at the exact multi-species steady drift solution, with the gas forced by the pressure-gradient mimic $2\Omega\eta v_K$. After 4495 cycles every deviation is at round-off:

	$\lvert\delta u_x\rvert$	$\lvert\delta u_\phi\rvert$	$\lvert\delta v_{x,1}\rvert$	$\lvert\delta v_{\phi,1}\rvert$	$\lvert\delta v_{x,2}\rvert$	$\lvert\delta v_{\phi,2}\rvert$	$\lvert\delta v_{x,3}\rvert$	$\lvert\delta v_{\phi,3}\rvert$
serial	1.3×10^{-16}	6.2×10^{-17}	6.9×10^{-17}	5.9×10^{-17}	1.2×10^{-16}	2.4×10^{-17}	9.4×10^{-17}	4.3×10^{-17}
4 ranks	1.1×10^{-16}	9.0×10^{-17}	1.2×10^{-16}	2.8×10^{-17}	1.3×10^{-16}	1.0×10^{-17}	9.0×10^{-17}	3.3×10^{-17}

This is the test that certifies the rotation/forcing kicks and the implicit drag are composed unsplit inside the IMEX stages: any splitting error would show as secular drift.

Damped dusty sound wave (Laibe & Price 2011; Krapp et al. §3.2). The exact complex eigenmode of the linearized gas–dust system, L_1 errors of density and velocity against the analytic damped wave at $t = 2$:

N_x	cycles	$L_1(\delta\rho)$	$L_1(\delta v_x)$	order
32	320	1.836×10^{-8}	1.351×10^{-8}	—
64	321	6.896×10^{-9}	5.124×10^{-9}	1.41
128	641	1.837×10^{-9}	1.377×10^{-9}	1.91
256	1281	4.765×10^{-10}	3.572×10^{-10}	1.95

The code's own root of the dispersion relation, $\omega = 4.529763 - 0.492158\,i$, is printed identically in every run. The order approaches two from below; the $32 \to 64$ step is polluted by the fixed-`cfl` time step at the coarsest resolution (the same cycle count at 32 and 64 cells), as the dust-tests notebook also observes.

3.2 Regressions

The ion-neutral C-shock (`inputs/ion-neutral/perp_cshock.athinput`, `integrator = imex2+`, 500 cycles) is the direct consumer of the refactored `imex2+` coefficient code: RMS-$L_1 = 2.071216 \times 10^{-3}$, bit-identical to the dust fork's baseline. The Sod tube runs through the untouched hydro task list.

3.3 Rank independence, and the deadlock

The damping problem with drifting particles (`v0 = 0.5`, $t_{s,3} = 0.5$, three lattice species) sends particles across MeshBlock and rank boundaries many times per run and exercises per-stage migration, the additive deposit exchange, the u^* ghost fill and the back-reaction exchange.

The first attempt at 4 ranks **hung at cycle 0**. The cause was in the merge itself: the dust fork's additive deposit exchange posted per-neighbor MPI messages against the base class of July's upstream, while this tree carries upstream #775, whose base-class receive posting is rank-packed (one aggregated message per peer rank). The sends had no receiver. After moving the exchange onto the rank-packed path (implementation document §4.4) with the serial path untouched (the damping ladder is bit-identical before and after the edit):

ranks	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\mathrm{tot}}\rvert$
1	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	5.2×10^{-15}
2	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	3.7×10^{-15}
4	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	3.6×10^{-15}
reference	1.158578×10^{-5}	(dust tests: "1-rank and 4-rank identical")			

The tiny spread in $\lvert\Delta P\rvert$ is the reduction order of the diagnostic itself. The NSH run at 4 ranks is in the table of §3.1.

3.4 Gravity regression

test	metric	measured	documented
`mg_poisson_slab`, sheet profile (absolute, no mean subtraction)	max rel / L_2 rel	1.203474×10^{-12} / 3.078736×10^{-13}	1.2×10^{-12}
`ts41_sin3` (Tomida & Stone §4.1)	defect ratio per V-cycle	0.1254, 0.1249, 0.1249, 0.1249	0.125
`mg_poisson_shear_bndry`, shwave, $t_0 = 0.37$	max rel / L_2 rel	5.950118×10^{-3} / 2.057024×10^{-3}	5.95×10^{-3} / 2.057×10^{-3}
same, $t_0 = 0$, shear-periodic vs periodic-x_1 twin	both runs	5.506611×10^{-3} / 1.281655×10^{-3}, identical	bit-identical

At 4 ranks the slab and boundary-annuli solves reproduce the serial digits exactly, as documented for Phases 2b and 3.

The dust module is inert without a `<dust>` block by construction (the hydro and gravity paths do not see it), and these numbers confirm the shared infrastructure it touched — driver coefficients, boundary-values base class, output tables, physics registration — left the solver untouched.

3.5 The integrator twin

The Phase-3 small stratified gravito-turbulence box (`inputs/shearing_box/gravito_turb_amr_test.athinput`: $16H \times 16H \times 12H$, $64 \times 64 \times 48$, $\beta = 10$, slab-open multigrid gravity, refinement switched off) evolved to $t = 3\,\Omega^{-1}$ with `rk2` and with `imex2+`, both at `cfl_number = 0.3`. Relative differences of the history columns at $t = 3$:

	mass	$E_{\rm tot}$	$E_{\rm int}$	$\langle \rho\, c_s \rangle$	$\langle \rho\, w_{\rm grv} \rangle$	$\langle \rho\, w_{\rm rey} \rangle$	$\langle \rho\, \delta v^2 \rangle$
$\lvert$imex2+ $-$ rk2$\rvert$/$\lvert$rk2$\rvert$	2.0×10^{-6}	7.0×10^{-5}	6.3×10^{-5}	3.5×10^{-5}	5.3×10^{-3}	3.2×10^{-4}	3.2×10^{-4}

Two different second-order integrators (three explicit stages of which the first advances by $\gamma \Delta t \approx 1.71 \Delta t$, versus two) on a chaotic flow: agreement at 10^{-4}–10^{-3} in the bulk quantities, with the small gravitational stress — a difference of large terms at this early time — the loosest at 5×10^{-3} (Fig. 2). The spikes of the relative difference of the stress columns at $t \approx 1.75$ and 2.5 in the figure are zero crossings of the stresses themselves (top-left panel), where a relative measure carries no information. This is the level at which `rk2`-validated gravity results transfer to dust runs; the full-size SC14 statistics under `imex2+` are a cluster item and will be attached to the next β-scan submission.

4 The second merge: regression, and the PC2/hybrid couplings

The dust fork's `dust` branch had moved on by seven commits (to `1f3e1ed9`) while the first merge was based on `d6de1296`; commit `78320aee` merges them (implementation document §4.5). They bring the fork's own rank-packed deposit exchange (taken in place of ours), CPU particle-mesh optimizations, and two new coupling modes selectable with `<dust>/coupling`: `pc2`, a per-cycle predictor–corrector particle update under `rk2`, and `hybrid`, which switches between PC2 and a first-order split backward-Euler branch on stiffness measures $\zeta = \max \Delta t / t_s$ and $\chi = \max \Delta t \rho_d / (\rho_g t_s)$.

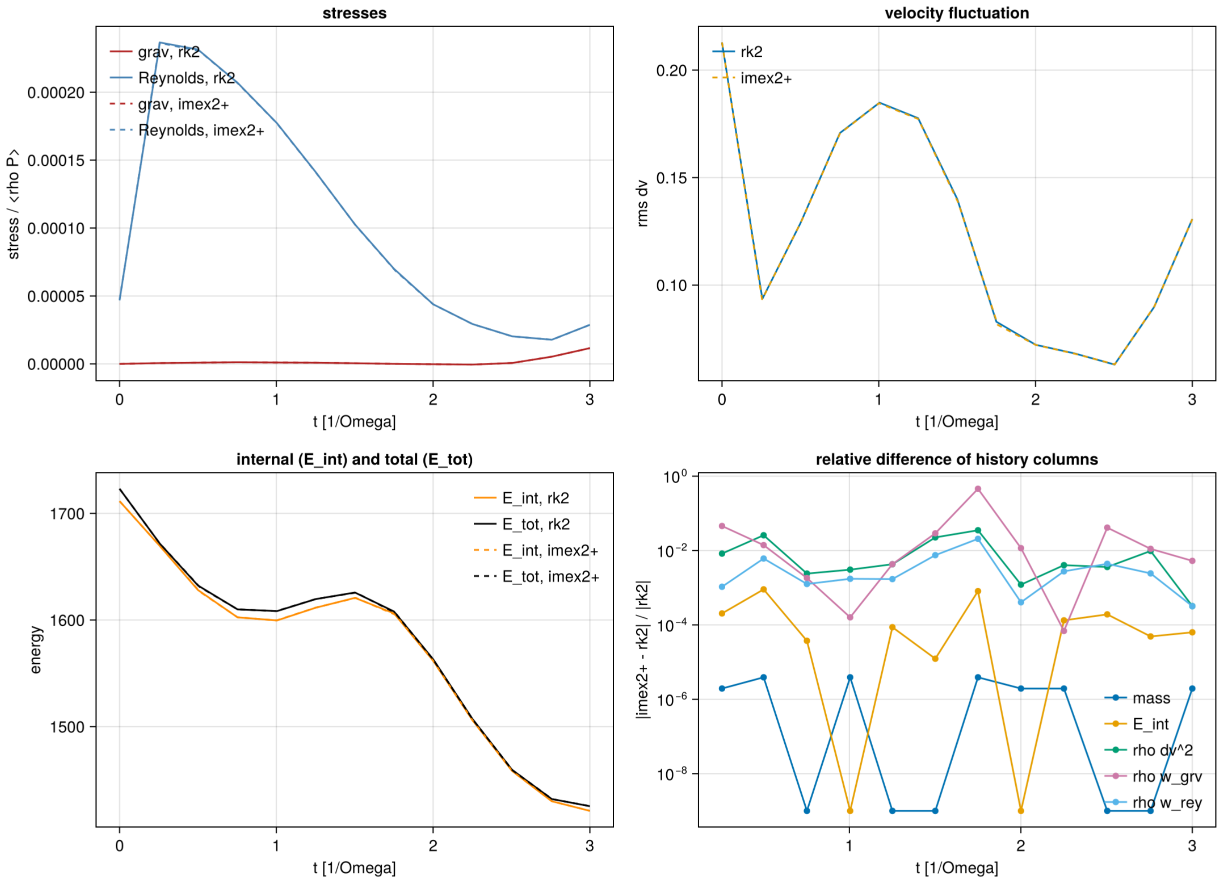

Figure 2: The gas-only gravito-turbulence box under `rk2` (solid) and `imex2+` (dashed): stresses normalized by $\langle \rho P \rangle$, rms velocity fluctuation, internal and total energy, and the relative difference of the history columns.

4.1 Regression

Every table of §3 was rerun on the rebuilt serial and MPI binaries. The damping ladder, its three layout variants, the dusty-wave ladder, the C-shock, the drifting-particle test at 1, 2 and 4 ranks, and all multigrid statics (serial and 4 ranks) reproduce the printed digits above exactly; the NSH deviations move within their 10^{-16} round-off (the new non-atomic deposit changes the summation order); and the gas-only gravito-turbulence run is **bit-identical** to the pre-merge history, as it must be — the merge touched nothing on the gas path.

4.2 PC2 coupling under `rk2`

The inputs are the same test inputs with `integrator = rk2` and `coupling = pc2` added to the `<dust>` block (archived under `run/dust4a/merge2/`; the key has to be in the file because the command line cannot add parameters). Damping ladder:

Δt	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\rm tot}\rvert$
1.235×10^{-2}	2.4349×10^{-7}	2.2596×10^{-7}	2.4748×10^{-7}	1.2039×10^{-6}	2.4×10^{-16}
6.211×10^{-3}	5.2577×10^{-8}	4.8577×10^{-8}	5.3355×10^{-8}	2.5966×10^{-7}	4.5×10^{-17}
3.115×10^{-3}	1.2351×10^{-8}	1.1216×10^{-8}	1.2461×10^{-8}	6.0729×10^{-8}	5.0×10^{-17}
1.558×10^{-3}	2.9941×10^{-9}	2.7081×10^{-9}	3.0167×10^{-9}	1.4707×10^{-8}	5.1×10^{-17}
order of err_u	2.23, 2.10, 2.04	(IMEX ladder: 1.98, 1.99, 1.99)			

Second order, converging onto the Δt^2 slope from above, with errors about 1.4–1.8× the IMEX ones at equal Δt and momentum at round-off (Fig. 1, open markers). The dusty wave:

N_x	$L_1(\delta\rho)$	$L_1(\delta v_x)$	order	(IMEX order)
32	1.612×10^{-8}	1.182×10^{-8}	—	—
64	4.483×10^{-9}	3.310×10^{-9}	1.85	1.41
128	1.139×10^{-9}	8.404×10^{-10}	1.98	1.91
256	2.858×10^{-10}	2.109×10^{-10}	2.00	1.95

Cleaner second order than the IMEX ladder and smaller errors at every resolution. The NSH equilibrium under `pc2` holds to round-off over the ten orbits (3.2×10^{-16}, 1.3×10^{-16}, 3.1×10^{-16}, 2.1×10^{-16}, 3.6×10^{-16}, 2.4×10^{-16}, 1.0×10^{-17}, 1.7×10^{-16} for the eight components of §3.1). Rank independence with the drifting particles:

ranks	err_u	err_{v1}	err_{v2}	err_{v3}	$\lvert\Delta P_{\rm tot}\rvert$
1	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	1.5×10^{-14}
2	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	1.9×10^{-15}
4	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	3.9×10^{-15}

4.3 Hybrid coupling and the split-BE branch

Two runs characterize the fallback. The three-species NSH test under `coupling = hybrid` (automatic selection) reports `pc2_cycles=0 split_be_cycles=4494` with $\zeta = 0.549$: the $t_s = 0.01$ species puts $\Delta t/t_s$ just above the 0.5 entry threshold, so the run stays in the first-order branch throughout, and the equilibrium is no longer an exact fixed point — deviations of $1\text{–}12 \times 10^{-5}$ after ten orbits instead of round-off. This is the designed behavior, not a defect: the first-order splitting error of a stiff fallback, on a problem whose IMEX and PC2 answers are exact. The stiff damping case of the dust tests ($t_s = 10^{-4}, 10^{-3}, 2\times10^{-3}$, $\Delta t/t_s$ up to 125) under `hybrid` runs split-BE for all 81 cycles ($\zeta = 32.8$, $\chi = 12.6$) and lands on the exact relaxed state (errors $\leq 4.9\times10^{-16}$, momentum 1.1×10^{-16}), the L-stability that the branch exists for. For production runs in the resolved regime the recommendation is therefore `coupling = pc2`; `hybrid` is the safety net for configurations with stopping times below the timestep.

5 Phase 4b (i): the shear-periodic deposit fold

The mechanism is in the implementation document §5: the x_1 ghost slabs of the shear-face MeshBlocks are gathered into global y–z planes, remapped in y by $\mp y_{\rm sh}$ (the adjoint of the ghost-fill signs), and added into the active edge strips across the face, while the unsheared plain-periodic x_1 buffers of those blocks are skipped. Data: `run/dust4b/` (README there).

5.1 The unit test

`pgen_name = dust_deposit_shear` (`inputs/dust/dust_deposit_shear.athinput`: $32 \times 32 \times 8$, $16 \times 16 \times 8$ blocks so both x_1 columns are face blocks, `nghost = 3`, $q = 3/2$, $\Omega = 1$, $L_x = 1$). Three fields live only in the x_1 ghost slabs of the face blocks: a smooth periodic g on the slab rows within the block's own y/z range, a constant on the same rows, and a constant on the whole slab including the corner ghosts (whose expectation is the analytic overlap multiplicity).

case	$y_{\rm sh}/\Delta y$	max rel (g)	const	multipl.	leak	mass
plain pass only (`fold = false`)	0	0	0	0	0	(none delivered)
fold, `time0 = 0`	0	0	0	0	0	5.9×10^{-16}
fold, 3-cell integer shift	3	3.5×10^{-16}	0	0	0	5.9×10^{-16}
fold, generic phase, 4 = 2 = 1 ranks	35.52	1.332091×10^{-3}	0	(frac.)	0	10^{-15}

The first row is the suppression check: with the fold switched off the receiving strips stay *exactly* zero, so no unsheared x_1 contribution survives (the first version of the direction table failed this row for the x_3x_1 corners; implementation document §5.4). Rows two and three are the exactness checks, including the multiplicity of overlapping corner rows. At a fractional shift the multiplicity field is a step profile and carries an $O(1)$ remap error at block boundaries, as expected; the smooth field's error is the remap interpolation error, which the last row shows to be identical at 1, 2 and 4 ranks to every printed digit.

Convergence. At a fixed fractional shift ($y_{\rm sh}/\Delta y = n_y/2 + 1/2$, so `eps` $= 1/2$ at every resolution; the generic-phase scan is confounded by the changing fraction):

n_y	DC L_1	DC max	PLM L_1	PLM max	PPMX L_1	PPMX max
32	1.86×10^{-3}	1.24×10^{-2}	2.69×10^{-4}	6.40×10^{-3}	1.11×10^{-4}	2.11×10^{-3}
64	4.66×10^{-4}	3.18×10^{-3}	3.89×10^{-5}	1.33×10^{-3}	9.16×10^{-6}	4.76×10^{-4}
128	1.17×10^{-4}	7.96×10^{-4}	5.33×10^{-6}	6.49×10^{-4}	9.90×10^{-7}	1.17×10^{-4}
256	2.91×10^{-5}	1.99×10^{-4}	6.21×10^{-7}	1.47×10^{-4}	9.78×10^{-8}	2.92×10^{-5}
order	2.00	2.0	2.8–3.1	∼2	3.2–3.6	2.0

Donor cell is exactly second order (the half-cell shift is symmetric); PLM, the default (matching the u^* remap), is second order in the maximum norm and better than second in L_1; PPMX is third order in L_1 with a limiter-bound maximum (Fig. 3). Mass is conserved to $\leq 3 \times 10^{-15}$ and the leak is exactly zero in every run.

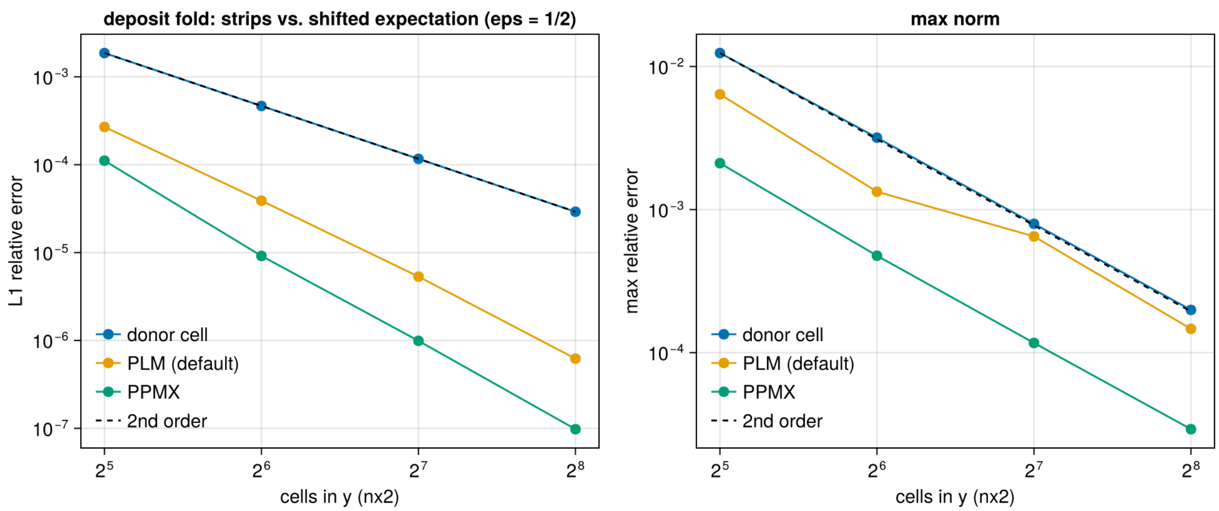

Figure 3: Convergence of the shear-periodic deposit fold at a fixed half-cell fractional shift, for the three remap orders.

5.2 Regression: the 3D shear-periodic NSH equilibrium

The new tracked input `inputs/dust/dust_nsh_shear3d.athinput` ($32 \times 32 \times 8$, $2 \times 2 \times 1$ blocks, ten orbits) is the azimuthally uniform state for which the old unsheared exchange was already exact:

	max deviation, before the fold	with the fold	PC2, with the fold	4 ranks
NSH, 3D shear-periodic	2.7×10^{-16}	2.6×10^{-16}	1.8×10^{-15}	1.1×10^{-16}

5.3 A structured run: rank independence and the size of the correction

Random particle placement in the NSH generator is rank-dependent by construction (it hashes tags to *local* block indices), which made a first serial-vs-MPI comparison look like a defect; with test particles the same runs agreed to 10^{-13}, and the cycle-0 dust density already differed. The deterministic alternative is the lattice with a 50% sinusoidal azimuthal modulation of the particle masses (`problem/mass_mod_amp = 0.5`, `check_equilibrium = false`), IMEX coupling with back-reaction, one orbit. Relative differences of the gas x-kinetic energy and of the species-1 radial-velocity sum:

t	serial vs 2 ranks	serial vs 4 ranks	fold vs. no fold	PLM vs. PPMX remap
0.81	10^{-14} / 7×10^{-15}	5×10^{-15} / 9×10^{-15}	4.2×10^{-2} / 2.7×10^{-2}	3.7×10^{-7} / 7.9×10^{-8}
3.20	1.2×10^{-14} / 5×10^{-15}	1.2×10^{-14} / 9×10^{-15}	9.7×10^{-3} / 5.8×10^{-3}	5.4×10^{-9} / 2.1×10^{-8}
4.80	6×10^{-15} / 7×10^{-15}	6×10^{-15} / 5×10^{-15}	6.8×10^{-2} / 3.9×10^{-2}	4.6×10^{-7} / 3.1×10^{-7}
6.28	6×10^{-15} / 5×10^{-15}	6×10^{-15} / 5×10^{-15}	5.4×10^{-3} / 3.6×10^{-3}	4.4×10^{-7} / 2.6×10^{-7}

Three conclusions. The fold is rank-independent to round-off on a structured field with back-reaction. The old unsheared exchange (`dust/shear_fold = false`) was a 1–7% error in the gas kinetic energy for a structured state, oscillating with the shear phase — not a small correction. And the remap order is a 10^{-7} effect on this run, so PLM is converged for the purpose.

6 Phase 4b (ii): the ideal-gas energy update

Mechanism: implementation document §6. Every drag kick on the gas momentum now books the matching kinetic-energy change (internal energy fixed under the kick); with `dust/drag_heating = true` the frictional dissipation of each particle kick is deposited as heat. Data: `run/dust4b/energy/`.

6.1 The damping test with an ideal gas

`inputs/dust/dust_damping_ideal.athinput`: the three-species damping problem of §3.1 with `eos = ideal`, $\gamma = 1.4$ and $p_{\rm gas} = 10^{-5}$ (comparable to the 1.4×10^{-5} of kinetic energy the drag dissipates, so the six-digit history resolves the budget; the first attempt with $p = 1$ could not), `dtmax` $= 1/95$ (the isothermal ladder's first step). The internal energy is $e = E - K_{\rm gas}$ from the hydro history, the dust kinetic energy $K_{\rm dust} = \sum_s \frac{1}{2}\epsilon_s\rho_0 V \bar{v}_s^2$ from the species velocity history (exact on the lattice).

Momentum dynamics. The four errors of the ODE test are identical between the default and heating modes to all printed digits (9.733252×10^{-8} etc. under IMEX, 1.660169×10^{-7} under PC2) — pressure and heat do not touch the velocities on a uniform state — and momentum is conserved to 10^{-16}.

Energy budgets versus the step. Errors relative to the dissipated kinetic energy, default mode ($|\Delta e|$) and heating mode ($|\Delta(E_{\rm gas}+K_{\rm dust})|$), with the order between successive rows in parentheses:

Δt	IMEX default	IMEX heating	PC2 default	PC2 heating
1/95	1.07×10^{-2}	3.81×10^{-2}	2.15×10^{-2}	5.89×10^{-2}
1/190	6.83×10^{-3} (0.65)	2.80×10^{-2} (0.44)	7.14×10^{-3} (1.59)	3.09×10^{-2} (0.93)
1/380	4.17×10^{-3} (0.71)	1.93×10^{-2} (0.54)	2.82×10^{-3} (1.34)	1.53×10^{-2} (1.01)
1/760	2.43×10^{-3} (0.78)	1.23×10^{-2} (0.65)	1.24×10^{-3} (1.18)	7.56×10^{-3} (1.02)

The default-mode residual is the register-combination gap $\gamma_0\gamma_1|\mathbf{K}_1|^2/2\rho$ per step (implementation document §6.3): the 1% at $\Delta t = 1/95$ is reproduced by that estimate with the stage-1 kick of the $t_s = 0.01$ species, and the sub-first-order IMEX slope is the saturation of that kick while $\Delta t \gtrsim t_s$ (the orders climb toward one as the step shrinks; PC2, with its RK2 weights, is already first order and better). The heating mode inherits the same gap on the dust side and converges at first order. Both are truncation errors of the tableau, not bookkeeping mistakes: the sign, size and Δt-scaling are the predicted ones.

6.2 The NSH state with an ideal gas

`inputs/dust/dust_nsh_shear3d_ideal.athinput` ($p = 1$, two orbits):

	max deviation of the drift equilibrium	$\dot{e}$ / analytic dissipation rate
default mode, `cfl 0.45`	4.6×10^{-15}	(no heat: e constant to 8×10^{-5})
heating, `cfl 0.45` ($\Delta t/t_{s,1} = 1.16$)	4.6×10^{-15}	1.0148
heating, `cfl 0.225`	—	1.0074
heating, `cfl 0.1125`	—	1.0037
heating, `cfl 0.45`, 4 ranks	6.7×10^{-16}	1.014777 (serial: 1.014777)

The analytic rate is $\sum_s \epsilon_s \rho_0 |\mathbf{u} - \mathbf{v}_s|^2/t_s$ from the equilibrium velocities. The heating rate converges to it at first order (the error halves with the step; the stiff $t_s = 0.01$ species dominates), the equilibrium is untouched by either mode, and 4 ranks reproduce the serial rate to every digit.

6.3 Regression

Isothermal runs cannot enter the new branches. The 2D damping ladder is bit-identical to the second-merge battery; the 2D and 3D NSH errors and the structured-run histories differ from their Phase-4b (i) values only in quantities at the 10^{-16} level (the deposit kernels changed shape and the compiler schedules the fused multiply-adds differently), and the 4-rank drifting-particle test reproduces the reference 1.158578×10^{-5} with the now four-channel exchange.

7 Phase 4b (iii): the particle timestep under shear

Mechanism: implementation document §7. The transport limit is now the peculiar velocity per cell (`dust/dt_transport = relative`, default) with a guard that no particle crosses more than `dt_block_safety` $= 0.5$ of a MeshBlock per stage on the total velocity $v_y - q\Omega x$; `dt_transport = full` is the legacy limit. Data: `run/dust4b/dt/`; figure 4.

7.1 Regression at the old step

The 2D damping ladder is bit-identical to Phase 4a (cycle counts 81, 161, 321, 642 and every error digit): the guard is inert without shear. In the $L_x = 1$ box of the tracked 3D input both particle limits lie above the gas limit (1.37×10^{-2} from c_s and $\Delta x_2 = 1/32$): legacy 2.1×10^{-2}, new guard 1.9×10^{-1}. The two modes therefore take the same 4588 steps and produce identical error lines (1.8×10^{-16}), PC2 its usual 6882 steps at 1.2×10^{-15} — which is also why the penalty had never shown in this test.

7.2 The wide box

The same input with $x_1 \in [-4, 4]$ at the same cell size (64×32×8, blocks 32×16×8, max $|q\Omega x| = 5.9$), ten orbits, uniform NSH state:

run	limit	Δt	cycles	CPU s	max NSH deviation
IMEX, serial	`full` (legacy)	2.60×10^{-3}	24043	455	8.8×10^{-16}
IMEX, serial	`relative`	1.37×10^{-2}	4588	94	1.3×10^{-15}
IMEX, 4 ranks	`relative`	1.37×10^{-2}	4588	30	1.1×10^{-16}
PC2/`rk2`, serial	`relative`	9.13×10^{-3}	6882	99	1.2×10^{-15}
IMEX ideal gas, 2 orbits	`relative`	1.37×10^{-2}	1082	—	5.4×10^{-15}

The legacy step is $0.5\,\Delta x_2/5.9$; the new step is the gas limit, the particle guard standing at $(0.5/1.707)\cdot 0.5/5.9 = 2.5\times10^{-2}$ above it. Every particle now hops MeshBlocks in y routinely (up to 4.4 cells per stage) and crosses the shear-periodic face with multi-cell azimuthal offsets; ten orbits pass without a halo violation, at round-off, on one rank and on four.

The guard. Lowering `dt_block_safety` until the guard is the active limit reproduces its formula $s/\beta_{\max}\cdot L_y^{\mathrm{mb}}/\max|v_y - q\Omega x|$ with $\beta_{\max} = 1.707$, $L_y^{\mathrm{mb}} = 0.5$ and $\max|v_y - q\Omega x| = 1.5\cdot 3.9375 + 0.027$:

s	0.1	0.2	0.4
formula	4.937×10^{-3}	9.873×10^{-3}	1.975×10^{-2}
measured Δt	4.936×10^{-3}	9.873×10^{-3}	1.370×10^{-2} (gas cap)

That the guard is needed, and not merely correct, is shown with thin MeshBlocks (`meshblock/nx2 = 4`, $L_y^{\mathrm{mb}} = 0.125$): at the gas step a particle near the radial edge moves $1.707\cdot 0.0137\cdot 5.9 = 0.14$ per stage, more than a block. With the guard the step drops to 6.12×10^{-3} and an orbit (1023 cycles) runs clean; with it disabled (`dt_block_safety = 4`, warned) the run aborts on the first cycle with the PM halo violation for a particle at $x = -3.81$, exactly the failure mode of §7.2 of the implementation document.

7.3 The structured run, and what the larger step costs

The azimuthally modulated state of §5 (`mass_mod_amp = 0.5`, back-reaction, one orbit) in the wide box. Rank independence first: the serial and 4-rank runs at the new step agree to 10^{-14} in every gas history column and to 5×10^{-14} in the species velocity sums, the same round-off as before at five times the step.

Between the two limits the histories differ visibly, and the difference is worth understanding rather than tabulating: for IMEX the gas x-kinetic energy at $\Delta t = 1.37\times10^{-2}$ differs from the legacy-step run by 5% at $t = 0.5$, 1% at $t = 3$ and 21% at $t = 6.28$, where the run goes through a streaming burst (figure 4, middle); for PC2 at its 9.1×10^{-3} the corresponding number is 1.5%. The step scan below (`<time>/dtmax`, the scratch input `run/dust4b/nsh3d_dtmax.athinput`) shows what this is:

gas x-KE	IMEX, $\Delta t =$ 1.37×10^{-2}	6.85×10^{-3}	3.43×10^{-3}	1.71×10^{-3}	orders
$t = 3$	1.40011×10^{-4}	1.38681×10^{-4}	1.38473×10^{-4}	1.38435×10^{-4}	2.7, 2.5
$t = 6.28$	2.6477×10^{-4}	2.2227×10^{-4}	2.1152×10^{-4}	2.0752×10^{-4}	2.0, 1.4
	PC2, $\Delta t =$ 9.1×10^{-3}	4.55×10^{-3}	2.28×10^{-3}		
$t = 3$	1.38623×10^{-4}	1.38550×10^{-4}	1.38496×10^{-4}		
$t = 6.28$	2.0991×10^{-4}	2.0835×10^{-4}	2.0658×10^{-4}		

The IMEX sequence converges (successive differences fall by 4–6× per halving; the order drops toward the end of the burst, where the state changes fastest), so the 21% is the IMEX coupling's truncation error at $\Delta t/t_{s,1} = 1.4$ — the stiffest species is under-resolved in time — and not an effect of the transport rule: the legacy limit had been supplying, by accident and only in wide boxes, a step of $t_{s,1}/4$. PC2 is closer to the converged value at every step (its 9.1×10^{-3} run is within 2% of the extrapolated IMEX limit, IMEX's own 2.6×10^{-3} run within 1.5%), consistent with the second-merge finding that PC2 is the production coupling in the resolved regime. The `gamma_switch` is inert here (it keys on the *largest* stopping time, $t_{s,3} = 1$). The practical rule: with `coupling = imex` and species stiffer than the gas step, cap the step with `<time>/dtmax` at a fraction of $t_{s,\min}$ if their dynamics matter; under PC2 the `rk2` step is adequate.

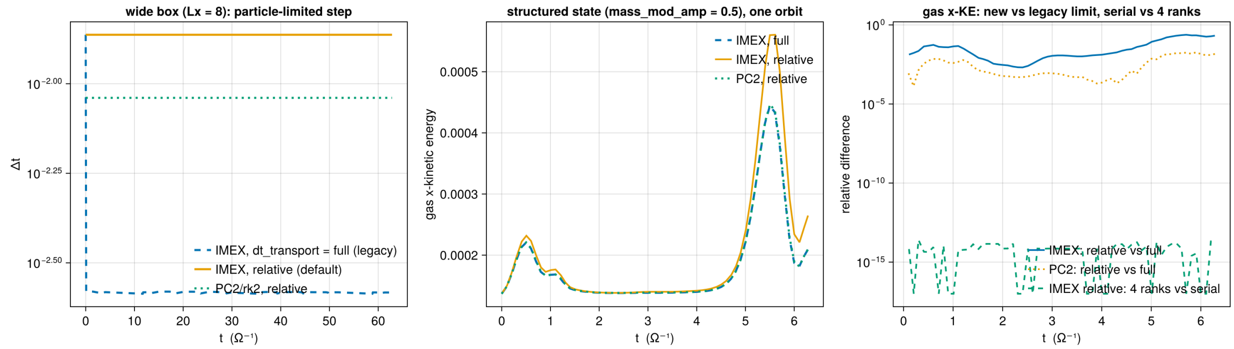

Figure 4: Phase 4b (iii). Left: the timestep of the wide-box NSH run under the legacy limit and the new default, IMEX and PC2. Middle: gas x-kinetic energy of the structured run over one orbit under both limits; the burst near $t = 5.5$ is where the IMEX truncation error at $\Delta t = 1.4\, t_{s,1}$ is largest. Right: relative differences of that history — new versus legacy step (truncation) and serial versus 4 ranks (round-off).

8 Phase 4c: dust self-gravity

Mechanism: implementation document §8. The particle-mesh dust density is added to the Poisson source, the mesh force $-\nabla\phi$ is gathered onto the particles with the deposit kernel, both solvers, the shear fold included. Data: `run/dust4c/` (README); figure 5.

8.1 The twin construction

A density profile with a known potential is split between the two components: one dust particle per cell at the cell centres carries a fraction f of the profile as mass ($m_p = f\rho V$), the gas keeps $(1-f)\rho$ (`<problem>/dust_frac` of the `fft_poisson` generator, `inputs/dust/dust_poisson_twin.athinput`). With NGP deposits the particle-mesh density is the profile to round-off, so every metric of the gas-only test must be reproduced exactly; with TSC the dust part is the $[\frac{1}{8}, \frac{3}{4}, \frac{1}{8}]^3$-filtered profile.

profile, solver	dust	Laplacian residual	analytic error	force
sin3 (TS23 4.1), MG 64^3	none (reference)	2.7×10^{-13}	rms 3.014475×10^{-5}	-
	NGP, $f = 1$	1.6×10^{-13}	3.014475×10^{-5}	8.004
	NGP, $f = 0.5$	2.7×10^{-13}	3.014475×10^{-5}	8.004
	NGP, $f = 1$, 4 ranks	1.6×10^{-13}	3.014475×10^{-5}	8.004
	NGP, $f = 1$, PC2	—	3.014475×10^{-5}	8.004
	TSC, $f = 1$ (serial, 4 ranks)	5.1×10^{-13}	1.052781×10^{-4}	7.970
sin3, FFT 64^3	none / NGP $f = 1$	$7.7/1.1 \times 10^{-14}$	3.014475×10^{-5}	— / 8.0
sheet, MG slab 64^3	none (reference)	1.4×10^{-12}	max 1.203474×10^{-12}	
	NGP, $f = 0.5$ (serial, 4 ranks)	1.4×10^{-12}	1.203474×10^{-12}	
	NGP, $f = 1$, gas floor 10^{-10} / 0		8.9×10^{-8} / 1.203474×10^{-12}	

The source is exact: every NGP twin reproduces the reference to the last printed digit, on both solvers, on 4 ranks (the additive exchange of the dust density across rank boundaries, and the slab Dirichlet planes gathered from the total density), and under the PC2 coupling. The one non-zero NGP entry is the gas density floor of the $f = 1$ slab run — the sheet test is absolute (no gauge), and a uniform 10^{-10} adds a parabolic potential at 10^{-7} of the sheet mode; with the floor at zero the digits return. The TSC twin's analytic error is the filter's, and its residual — the solve against its own (filtered) source — is at round-off.

The force. The acceleration gathered at the particles against the analytic $-\nabla\phi$ of the sin3 potential, maximum relative error over all particles:

cells per side	32	64	128	orders
NGP	3.169×10^{-3}	8.004×10^{-4}	2.006×10^{-4}	1.99, 2.00
TSC	3.115×10^{-2}	7.970×10^{-3}	2.004×10^{-3}	1.97, 1.99

NGP at cell centres returns the cell's centred-difference gradient, and its error is that operator's $O(\Delta x^2)$ truncation; TSC applies the three-cell filter twice (deposit and gather), ten times the error at the same order.

Momentum. With the dust carrying the Gaussian blob and the gas the modes of the `modes` profile (`dust_part = blob`), the two components have different shapes and exert a net force on each other. The sum of the net force on the gas ($\sum \rho_g \mathbf{g} V$, the gas source term's own operator) and on the dust ($\sum_p m_p \mathbf{g}(\mathbf{x}_p)$, the gather), relative to the net dust force:

$\lvert\mathbf{F}_g + \mathbf{F}_d\rvert/\lvert\mathbf{F}_d\rvert$	32	64	128
NGP	6.5×10^{-14}	2.7×10^{-13}	6.8×10^{-13}
TSC	8.4×10^{-14}	8.3×10^{-14}	4.7×10^{-13}

Round-off, not truncation, for both kernels: the identity of implementation §8.2 (one kernel for scatter and gather, an antisymmetric gradient of a symmetric Laplacian's solution) holds discretely, to the solver's convergence.

8.2 Test particles in a self-gravitating slab

`inputs/dust/dust_grav_orbit.athinput` (generator `dust_grav_orbit`): an isothermal gas slab $\rho_0\,\mathrm{sech}^2(z/h)$, $h = c_s/\sqrt{2\pi G\rho_0} = 0.141$ (9 cells), in hydrostatic equilibrium under its own gravity (multigrid, slab-open faces, `diode` hydro boundaries so nothing falls in), $32 \times 32 \times 64$ cells at $\Delta = 1/64$; 64 test particles (not in the source, no drag on the gas, stopping time 10^{10}) released at rest at $z_0 = 0.21\,h$. The exact period at this amplitude, from the quadrature of $\phi = 2c_s^2 \ln\cosh(z/h)$, is $T = 6.31832$; the harmonic period $2\pi/\sqrt{4\pi G\rho_0} = 6.28319$ is 0.56% shorter. Period from the sign changes of $\langle v_z\rangle$ over five periods:

	IMEX $\Delta t = 0.070$	0.050	0.025	0.0125	PC2 0.045
$T/T_{\mathrm{exact}} - 1$	$+8.1 \times 10^{-4}$	-2.5×10^{-4}	$+4.8 \times 10^{-5}$	$+1.2 \times 10^{-4}$	$+8.0 \times 10^{-5}$
amplitude after 5 periods	+1.4%	−0.36%	−0.37%	−0.37%	−0.42%
spread of z over particles	2×10^{-5}	2×10^{-9}	2×10^{-9}	2×10^{-9}	2×10^{-9}

The anharmonic correction is resolved, the period converges to the exact value to a floor of $\sim 10^{-4}$ set by the slab's own slow evolution (its $\langle \rho z^2 \rangle$ drifts by -0.23% over the run, and the amplitude follows it), and all 64 particles, scattered over the (x, y) plane, trace one trajectory to 10^{-9} — the force field of a z-only potential has no (x, y) imprint through the TSC gather. With NGP the period is 5.6% long: the piecewise-constant force of the nearest cell, the reason TSC is the default.

8.3 The NSH state with gravity

`inputs/dust/dust_nsh_shear3d_grav.athinput` adds a periodic multigrid solve to the 3D shear-periodic NSH box (reshaped to $32 \times 32 \times 16$ in $L_z = 0.5$: cubic cells and 16^3 blocks). The uniform lattice with the dust in the source and the particles feeling ϕ holds its equilibrium to 2.5×10^{-16} over an orbit (gravity inert: 2.3×10^{-16}) — a uniform total density is a constant potential through the fold, the ghost fill and the shear remap of both the density and the force. The structured state (mass modulation 0.5, no drift, $4\pi G = 2$) moves only through gravity: the gas x-kinetic energy rises to 5×10^{-5} at $t = 1.2$ as the pattern is sheared apart; serial and 4-rank runs agree in every kinetic-energy column to 10^{-14}, multigrid and FFT to 2×10^{-4} while the force is real ($t < 2.4$), IMEX and PC2 to 1.6% at the peak. The net momenta of this state vanish by symmetry (a sinusoidal pattern), which is why the antisymmetry is measured statically above.

8.4 Regression, and what was found

Without a `<gravity>` block nothing new executes: the 2D damping line and the one-orbit wide structured history of Phase 4b (iii) are bit-identical (cycle counts and every digit), before and after the `Particles::dtnew` fix.

The drag-coupling instability. Setting up the gravity box exposed that the 3D NSH lattice, round-off-exact at $32 \times 32 \times 8$, is not at $32 \times 32 \times 32$: the vertical kinetic energy grows from 10^{-37} at $t = 0.6$ to 10^{-5} at $t = 2.4$ ($1.6\times$ per step), and the equilibrium errors reach 10^{-6} by one orbit. The variants at $n_z = 32$ (`run/dust4c/inst/`):

stable to round-off	unstable	
$n_z = 16$; $n_z = 64$; $L_z = 2$ ($\Delta z = 2\Delta x$)	$n_z = 24, 32$ ($\Delta z \le \Delta x$)	geometry
CIC, NGP deposits	TSC	kernel
`imex2+` at `cfl` 0.225, 0.30, 0.35	`cfl` 0.40, 0.45	step
PC2/`rk2` at `cfl` 0.30	`cfl` 0.45, 0.55	step
$\eta v_K = 0$ (exactly zero)	drift on; also with $t_{s,1} = 0.1$; also without back-reaction	physics

The threshold is $c_s \Delta t / \Delta z \approx 0.35$ with the *cycle* step, the same for both integrators (so it is not the gas Courant condition of a stage), and it needs the TSC kernel and the drift. A growth rate of $35\,\Omega$ and the stability of CIC at the same step rule out the streaming instability. This is a property of the inherited drag coupling that the Phase-4b boxes ($\Delta z \ge 2\Delta x$) never reached; its mechanism is a Phase-4d item. The operating rule meanwhile — `cfl_number` ≤ 0.3 where $\Delta z \le \Delta x$ — is written into the gravity NSH input, and every Phase-4c result above was obtained under it.

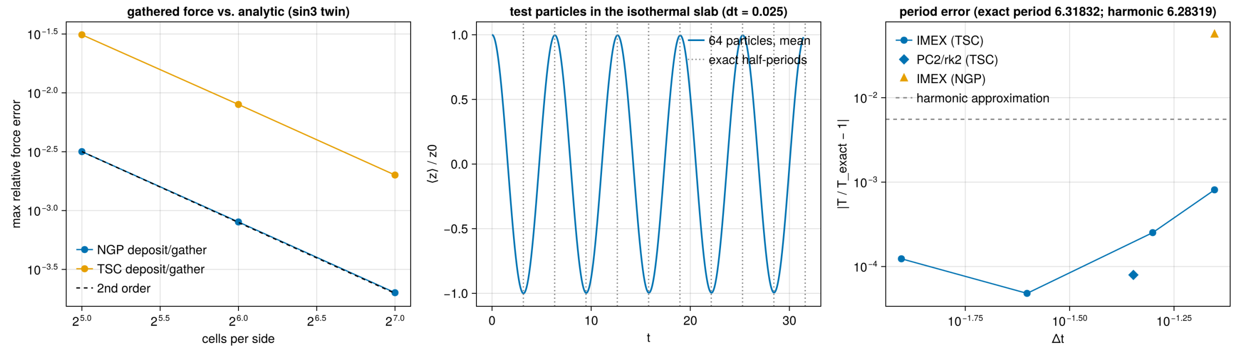

Figure 5: Phase 4c. Left: the force gathered at the particles against the analytic acceleration of the triple-sine potential, NGP and TSC, second order. Middle: the mean height of the 64 test particles in the isothermal slab at $\Delta t = 0.025$, with the exact half-periods marked. Right: the period error against the step; the harmonic approximation would sit at the dashed line, the NGP kernel at the triangle.

9 Phase 4d: the instability identified, the vertical policy, restart, diagnostics

Mechanism: implementation document §9. Data: `run/dust4d/` (README); figure 6.

9.1 The instability is the hydro scheme's

Binary outputs of the unstable 32^3 dust run of §8 (`cfl` 0.45, every 0.2 in time) give the growing mode directly: the power of the gas v_z, v_x and ρ concentrates at $(k_x, k_y, k_z) \approx (16, \pm 15, \pm 15)$ of 32 from $t = 1.2$ on, growing as e^{35t} between $t = 1.2$ and 2.0 (figure 6, right). The same box with no particles, gas velocities seeded with 10^{-10} white noise (the NSH generator's new `<problem>/gas_vpert`), one orbit:

gas only, 32^3, seeded	$t = 0.5$	$t = 2$	$t = 4$	end
`imex2+`, `cfl` 0.45	1.8×10^{-9}	1.8×10^{-5}	4.0×10^{-5}	1.7×10^{-5}
`rk2`, `cfl` 0.45	1.8×10^{-9}	3.5×10^{-5}	4.8×10^{-5}	3.1×10^{-5}
`rk2`, `cfl` 0.40	2.5×10^{-17}	1.2×10^{-5}	8.2×10^{-6}	5.6×10^{-6}
`imex2+`, `cfl` 0.30	4.6×10^{-24}	1.6×10^{-24}	5.8×10^{-25}	4.7×10^{-25}
`rk2`, `cfl` 0.30	4.8×10^{-24}	1.6×10^{-24}	5.9×10^{-25}	4.7×10^{-25}
`imex2+`, `cfl` 0.45, $n_z = 8$	7.3×10^{-24}	1.6×10^{-24}	1.9×10^{-25}	6.3×10^{-26}

(vertical kinetic energy; the seed is $\sim 10^{-20}$). With no dust in the box the mode grows at `cfl` 0.45 and 0.40 under both integrators, to the same saturation the dust runs reached, and decays at 0.30 and on the coarse-z mesh: the instability is the 3D shearing-box hydro's, with the dust's deposit round-off as the seed that the exactly uniform gas-only NSH never had. Every stable/unstable entry of the Phase-4c scan is reproduced by the checkerboard frequency $\omega = 2c_s\sqrt{\sum \Delta x_i^{-2}}$ with $\omega\Delta t \approx 1.2$–$1.3$ as the threshold. The rule `cfl` ≤ 0.3 in 3D stands, as a hydro rule.

9.2 Vertical outflow policy

`dust_grav_orbit` with `problem/vz0 = 0.3`, no gravitational force, in the diode-faced slab box: 64 particles launched at $v_z = 0.3$ from $z_0 = 0.03$ reach the face at $t \approx 1.6$; the history's particle count drops from 64 to 0 between $t = 1.51$ and 1.85, `DUST_ESCAPE_SUMMARY removed=64 mass=64`, and the run continues to $t = 3$ with no particles and no abort — serially, and on 2 ranks where the two species leave through opposite faces from different ranks. With every face periodic the marking never fires: the 2D damping ladder (81 cycles, 1.352906×10^{-7}) and the narrow 3D NSH orbit (459 cycles, 9.393528×10^{-16}) are bit-identical.

9.3 Particle restart

The structured 3D NSH box (`run/dust4d/rst/nsh_rst.athinput`: mass modulation 0.5, one orbit, restart dumps every 3.0): the run restarted from the $t = 3$ dump reproduces the uninterrupted run's gas and dust histories to every digit over the 33 common rows, serially and on 4 ranks.

9.4 The `dust_dpm` output

On the sin3 twin with $f = 1$ and NGP deposits, `dust_dpm` plus the (floored) gas density equals the gas-only reference density to 0.0; with TSC the two differ by the filter (3.6×10^{-3} maximum), with the mean exactly 2 in both. On the 3D shear-periodic NSH lattice ($\epsilon = 1$) the output is 1.000000000000000 in every cell, serially and on 4 ranks — the additive exchange and the fold are in it.

9.5 The two-stage science setup, exercised

The Baehr et al. runs (implementation §9.5) are a gas-only stage to saturation and a restart that inserts the particles. The mechanics were exercised in a 32^3 box ($L = 6.9$, $\Delta = 0.216 = 0.1\,H_g$, cooling off so the layer merely relaxes): stage 1 to $t = 0.5$ with a dump at 0.25; stage 2 restarted from that dump with the add-on input, back-reaction on and off, serially and on 4 ranks. The generator reports 736 particles inserted (target 732 from `ppc`), each of mass $ZM_{\rm gas}/N$, and the history's `d_mass` column reads $0.970981 = 0.01 \times 97.0981$ from the first row; the dust momenta and kinetic energy evolve, `d_escaped` stays zero (the layer is well inside the faces), the `dust_dpm` and `pvtk` outputs are written, the serial and 4-rank BR histories agree to 0.0, and BR and noBR diverge as they must. The particle history reads back the insertion: 736 particles, mass 0.97098, $\sigma_x = 2.04 = L/\sqrt{12}$ for the uniform (x, y) placement, $\sigma_z = 1.70$ for the $H_g = 2.17$ Gaussian truncated at ± 3.45, σ_{v_z} growing from 0.1 to 1.5 as the layer starts to settle; and the Table-1 script runs end to end on the same directory (all quantities produced; the physics of a $0.1/\Omega$ mini run is of course meaningless). The irradiation floor of the cooling (`bcool_cs2_floor`) was checked for side effects on the same box: with the floor, without it, and with cooling off, the 40 first steps have identical Δt histories. For the Vista estimate the archived SC14 box shows the step settling at 4×10^{-3} (the free-falling floor halo), which is what the job scripts assume.

10 Summary

Phase 4a delivers the dust module inside the multigrid tree with both sides certified unchanged: the dust fork's acceptance numbers and the gravity solver's documented statics reproduce to every printed digit, serially and at 4 ranks, and the one merge defect (the additive deposit exchange against rank-packed messaging) is fixed with a rank-independence table to show for it. The integrator the dust module's IMEX coupling imposes, `imex2+`, tracks `rk2` on the stratified gravito-turbulent box at the level two second-order schemes should — and after the second merge that constraint is optional:

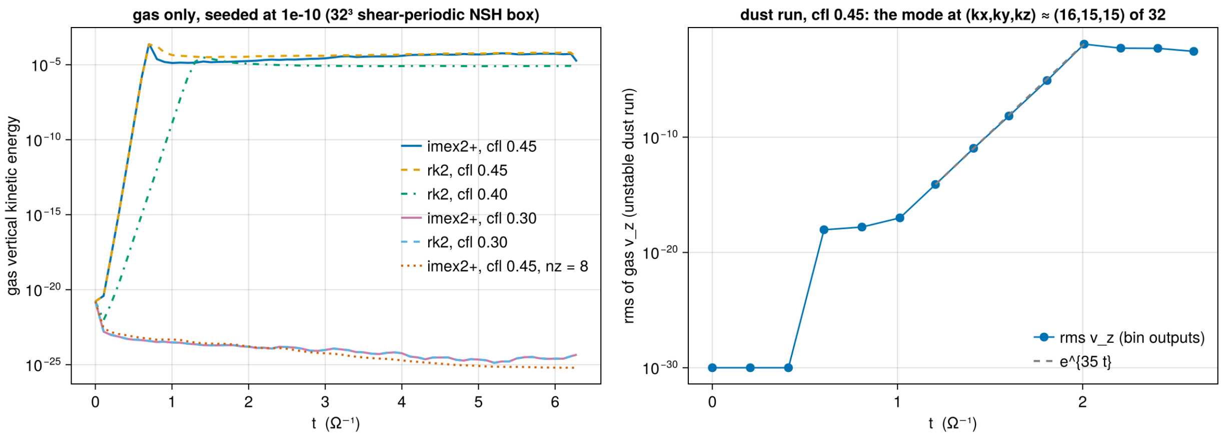

Figure 6: Phase 4d. Left: the gas-only 32^3 box seeded at 10^{-10}: the vertical kinetic energy grows at `cfl` ≥ 0.4 under both integrators and decays at 0.3 and at $n_z = 8$. Right: the rms gas v_z of the unstable dust run from its binary outputs, with the e^{35t} growth of the checkerboard mode.

the PC2 coupling runs under `rk2` at second order with the same conservation properties, and is the recommended production path in the resolved drag regime.

Phase 4b (i) closes the first gap Phase 4a left open: ghost deposits now cross the shear-periodic radial faces with a conservative azimuthal remap, certified exact at integer shifts, convergent otherwise, conservative, leak-free and rank-independent, and shown to matter at the percent level on structured states. Phase 4b (ii) makes back-reaction usable with an ideal gas: the energy bookkeeping is exact per kick, the residuals are the RK combination gap (a percent of the dissipated energy at a step equal to the stiffest stopping time, converging), and the optional frictional heating reproduces the analytic dissipation rate to first order. Phase 4b (iii) closes the third: the particle step no longer pays the full-shear-velocity penalty — 5.2× fewer cycles in an $L_x = 8$ box, with the equilibrium, rank independence and the MeshBlock guard each certified — and the scan it prompted fixes the operating rule for stiff species under IMEX (cap the step at a fraction of $t_{s,\min}$, or use PC2).

Phase 4c delivers gas+dust self-gravity: the source is certified exact by twin construction on both solvers, serially and on 4 ranks; the force converges at second order and conserves the total momentum to round-off by construction; test particles orbit in a self-gravitating slab with the exact anharmonic period; and the shear-periodic machinery (fold, ghost fill, remap) carries the dust density and the force without disturbing an equilibrium. The dust track now has every ingredient of the target problem except the vertical boundary policy and the diagnostics of Phase 4d.

Phase 4d supplies those and closes the one open numerical question: the instability is the hydro scheme's 3D checkerboard mode at `cfl` $\gtrsim 0.4$, not the dust coupling's, so the Courant rule is understood; particles leave through the vertical faces with their mass accounted; restart chains carry the particles exactly; and the module's own dust density and the SC14 dust history columns are available. The dust track is ready for the science box, and the two-stage Baehr et al. setup — the gas to saturation, then the particles inserted on restart, back-reaction a switch — is built, exercised in a small box, and scripted for Vista.